

Response to Reviewers' Comments

Submission 46: “Low-Resource ASR for Kalenjin: Two-Stage Fine-Tuning of Wav2Vec2-XLS-R with KenLM Integration”

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Dear Editors and Reviewers,

We sincerely thank the reviewers for their careful reading of our manuscript and their constructive feedback. The comments have significantly strengthened the quality and rigor of our paper. Below, we provide a detailed point-by-point response to each comment, clearly describing the revisions made to address the concerns raised.

All modifications are reflected in the revised manuscript. New additions include Table VII (Ablation Study), Table VIII (Error Type Distribution), an expanded Error Analysis subsection, a Generalizability discussion, and enhanced dataset documentation.

Response to Reviewer 1

Overall Score: 8.1/10 Decision: Accept

We are grateful to Reviewer 1 for the thorough and encouraging evaluation. We address each point below.

1. Novelty & Contribution (8.5/10)

Reviewer Comment: *The combination of cross-lingual transfer (Wav2Vec2-XLS-R), staged fine-tuning, and lightweight language modeling is not entirely novel individually, but its application and systematic evaluation in this context is valuable and impactful.*

Our Response: We appreciate this assessment. We agree that the individual techniques are established; our contribution lies in their systematic application to an undocumented language, the identification of training pathologies specific to extremely low-resource settings, and establishing the first ASR baseline for Kalenjin. No changes were required for this point.

2. Technical Soundness (8/10)

Reviewer Comment: *The final WER (61.75%) remains relatively high, which is expected but could benefit from deeper error analysis or qualitative insights.*

Our Response: We have added a comprehensive error analysis based on 200 randomly selected test utterances (Section V, new subsection “Error Analysis”). We systematically categorized 591 error instances into five types:

- Rare/OOV vocabulary: 51.9% (307 errors)
- Insertions: 17.4% (103 errors)
- Phonetic confusion (c/ch, g/k pairs): 13.4% (79 errors)
- Deletion of function words: 12.5% (74 errors)
- Morpheme boundary errors: 4.7% (28 errors)

This analysis reveals that the dominant error source is insufficient training vocabulary coverage (18,576 unique words), with over half of all errors involving word forms unseen or underrepresented during training. This provides concrete direction for future improvements through expanded text corpora. The results are presented in the new Table VIII.

3. Methodological Clarity (9/10)

Reviewer Comment: *Descriptions are precise, especially regarding language-specific normalization. The discussion of training pathologies is particularly strong.*

Our Response: We thank the reviewer for this positive assessment. No changes were required.

4. Experimental Rigor (7.5/10)

Reviewer Comment: *The study would benefit from: (a) Comparisons with alternative architectures or baselines (if feasible); (b) More detailed ablation studies (e.g., impact of preprocessing components or LM integration individually).*

Our Response: We have addressed both concerns:

- Baseline comparisons:** The revised manuscript now includes an explicit comparison table (Table VI) contextualizing our results against the PazaBench benchmark, which evaluates Omnilingual, Paza, Facebook MMS, Whisper, Phi-4, and other models on Kalenjin. Our fine-tuned system achieves CER 50% lower than the best zero-shot model (Omnilingual) and WER 23.5% lower.
- Ablation study:** We have added Table VII presenting an ablation analysis with the following configurations:
 - Stage 1 (frozen encoder): WER 72.11%, CER 44.94%
 - Stage 1 (batch optimized): WER 69.03%
 - Stage 2 (unfrozen encoder): WER 69.13%, CER 20.11%
 - Stage 2 greedy decoding: WER 73.64%, CER 18.33% (50-sample subset)
 - Stage 2 + KenLM beam search: WER 60.00%, CER 16.14% (50-sample subset)
 - Full test set + KenLM: WER 61.75%

The ablation demonstrates that language model integration provides the largest single-component improvement (13.64 percentage points on the evaluation subset), while Stage 2 acoustic adaptation provides the most significant CER improvement (55.3% relative reduction).

5. Results & Impact (8/10)

Reviewer Comment: *The 55.3% relative CER reduction is significant. The additional 10.7% WER improvement from KenLM reinforces the importance of language modeling.*

Our Response: We thank the reviewer for recognizing the significance of these results. The ablation study now further reinforces this finding by showing the LM contribution in isolation.

6. Writing Quality & Organization (7.5/10)

Reviewer Comment: *Some sentences are overly long. Transitions between components can be smoother. Minor redundancy and punctuation issues.*

Our Response: We have revised the manuscript to address these concerns:

- Shortened overly long sentences throughout, particularly in the Key Findings subsection
- Removed redundant qualifiers and filler words (e.g., “dramatic”, “substantially”)
- Improved transitions between subsections in the Discussion
- Fixed spacing around parentheses and punctuation inconsistencies

Response to Reviewer 2

Decision: Accept (with moderate revisions)

We sincerely thank Reviewer 2 for the supportive evaluation and the constructive suggestions for improvement. We address each point below.

1. Dataset Description

Reviewer Comment: *The dataset description requires more detail regarding speaker diversity, recording conditions, and preprocessing procedures to support reproducibility.*

Our Response: We have expanded Section III-A (Data Acquisition and Corpus Characteristics) to include:

- **Speaker diversity:** The corpus includes contributions from approximately 150 unique speakers with varying age groups, genders, and dialectal backgrounds across Kalenjin sub-varieties.
- **Recording conditions:** Audio was recorded via participants’ personal devices (primarily smartphones and laptops) in uncontrolled acoustic environments including homes, offices, and outdoor settings, through the Mozilla Common Voice web platform.
- **Speaker-disjoint evaluation:** We explicitly confirm that no speaker appears in more than one split, ensuring unbiased evaluation.
- **Preprocessing:** The complete preprocessing pipeline is documented in Sections III-B through III-D, with quality thresholds in Table IV and rejection statistics (0.09% rejection rate).

2. Comparative Analysis

Reviewer Comment: *The comparative analysis with baseline models and related low-resource ASR approaches could be expanded further.*

Our Response: The revised manuscript now includes:

- Table I: PazaBench benchmark results for 8 models on Kalenjin (Omnilingual, Paza, Facebook MMS, Wav2Vec2-Conformer, Data2Vec, Whisper, Phi-4, Moonshine)
- Table VI: Direct comparison of our fine-tuned system against PazaBench baselines
- Discussion of why zero-shot models (Whisper WER 1.00, MMS WER 1.03) fail on Kalenjin while targeted fine-tuning succeeds
- Table VII: Ablation study isolating individual component contributions

3. Error Analysis and Limitations

Reviewer Comment: *The discussion section would benefit from a deeper analysis of errors, limitations, and generalizability to other underrepresented African languages.*

Our Response: We have added three new subsections to the Discussion:

1. **Error Analysis (new):** Systematic categorization of 591 errors from 200 test utterances into five types (Table VIII), with analysis of root causes and implications for future work.
2. **Generalizability to Other African Languages (new):** Discussion of how the methodology applies to other under-resourced African languages, particularly Nilotic family languages (Dholuo, Maasai, Turkana) that share tonal phonology and agglutinative morphology. We identify the key transferable components: quality filtering framework, staged unfreezing strategy, and lightweight KenLM integration.
3. **Limitations (expanded):** Added acknowledgment of limited speaker metadata, the absence of demographic bias analysis, and the constraint of the LM to 17,814 unigrams from training transcriptions only.

4. Grammar and Formatting

Reviewer Comment: *Minor grammatical and formatting inconsistencies should also be revised before final publication.*

Our Response: We have carefully proofread the entire manuscript and corrected:

- Grammatical inconsistencies and punctuation errors
- Formatting of tables and figure captions for consistency
- Spacing issues around mathematical notation
- Consistent use of terminology throughout (e.g., “under-resourced” vs. “low-resource”)

Reviewer Concern	Action Taken
Deeper error analysis	Added Table VIII: 591 errors categorized across 200 test utterances
Ablation studies	Added Table VII: component-level ablation with real experimental data
Dataset details	Expanded Section III-A with speaker diversity, recording conditions
Comparative baselines	Added PazaBench comparison (Table I, Table VI)
Generalizability	New subsection on applicability to other African languages
Writing quality	Shortened sentences, improved transitions, fixed punctuation
Limitations	Expanded with speaker metadata, demographic bias, LM coverage

Summary of Revisions

We believe these revisions comprehensively address all reviewer concerns and significantly strengthen the manuscript. We are grateful for the opportunity to improve our work and look forward to the reviewers' assessment of the revised version.

Sincerely,

Kevin Omondi Obote, Annah Mumbua Muli, Vivian Nyamari Moraa, and Joyce Muthoni Njiiri